

Mini Project: Rule-Based Classifier

This is a small project I made for practice. The idea is to build a simple rule-based classifier to predict whether a student will Pass or Fail in an exam. I created a dataset with some basic features and then used simple if-else conditions to make predictions.

Features used in the dataset:

1. Study Hours (number of hours studied)
2. Attendance (High/Low)
3. Assignments Completed (Yes/No)
4. Result (Pass/Fail)

Sample Dataset (10 rows):

Study_Hours	Attendance	Assignments_Completed	Result
8	High	Yes	Pass
5	High	Yes	Pass
2	Low	No	Fail
6	High	No	Pass
1	Low	No	Fail
7	High	Yes	Pass
3	Low	Yes	Fail
4	High	No	Pass
2	Low	No	Fail
9	High	Yes	Pass

Rules I used for classification:

- If Study_Hours < 3 and Assignments_Completed = No → Fail
- Else if Attendance = Low and Study_Hours < 5 → Fail
- Else → Pass

Python Code:

```
def predict_result(study_hours, attendance, assignments):  
    if study_hours < 3 and assignments == "No":  
        return "Fail"  
    elif attendance == "Low" and study_hours < 5:  
        return "Fail"  
    else:  
        return "Pass"
```

```
# Testing the function
```

```
dataset = [  
    [8, "High", "Yes"],  
    [2, "Low", "No"],  
    [6, "High", "No"],  
    [1, "Low", "No"],  
    [4, "High", "No"]  
]
```

```
for row in dataset:
```

```
    print(row, "->", predict_result(*row))
```

This project was mainly for learning purposes. It shows how simple rules can be used to classify data without using any complex machine learning algorithms.